

COSC 594

DM Alignment (I)

Dongsheng Ding

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

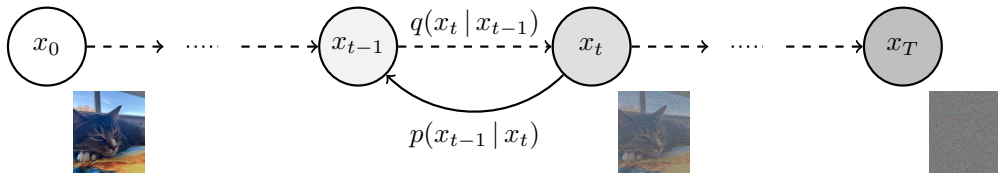
Outline

- RECAP
- ALIGNMENT OF DM
- REWARD-BASED ALIGNMENT

Outline

■ RECAP

Diffusion model



data point x_0 – image, trajectory, video, protein

■ ANALOGUE W/ VAE

- ★ input and latent

$$x = x_0 \quad \text{and} \quad z_{1:T} = x_{1:T}$$

- ★ noiser (encoder)

$$q(x_{1:T} | x_0) = q(x_1 | x_0)q(x_2 | x_1) \dots q(x_T | x_{T-1})$$

- ★ denoiser (decoder)

$$p(x_0 | x_{1:T}) = p(x_T)p(x_{T-1} | x_T) \dots p(x_0 | x_1)$$

■ GROUND-TRUTH DENOISER

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma_q^2(t)I)$$

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}x_0 - \text{mean}$$

$$\sigma_q^2(t) = \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} - \text{variance}$$

■ APPROXIMATE DENOISER

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_q^2(t)I)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\mu_q(x_t, x_0) - \mu_\theta(x_t, t)\|^2$$

$$\text{Pretrained model } p_t^{\text{pre}}(x_{t-1} | x_t) = p_{\theta^\star}(x_{t-1} | x_t)$$

Outline

- RECAP

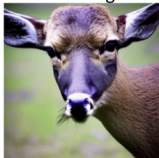
- ALIGNMENT OF DM

Downstream requirements

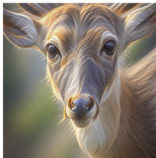
■ AESTHETIC QUALITY

Prompt: *A photo of a deer*

No fine-tuning

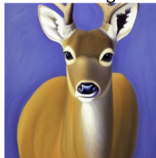


Aesthetic reward

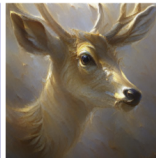


Prompt: *A painting of a deer*

No fine-tuning



Aesthetic reward



■ HUMAN PREFERENCE



a painting of an old man



a painting of a racoon
washing dishes

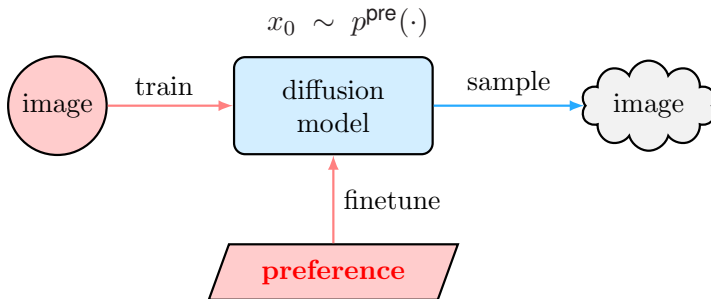
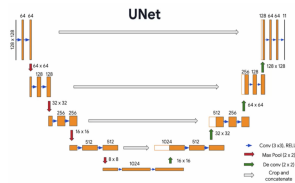


a painting of a tree



a painting of a city skyline

DM alignment framework



preference – reward / human value

Reward-based alignment

Step 1: Diffusion Model Pretraining



Step 2: Reward Modeling



Step 3: Reinforcement Learning-Based Fine-Tuning

Outline

- RECAP
- ALIGNMENT OF DM
- REWARD-BASED ALIGNMENT

Diffusion model sampling

Pretrained diffusion model: $p_t^{\text{pre}}(x_{t-1} | x_t)$ for $t = T, \dots, 2, 1$

■ MARKOV CHAIN

$$\underbrace{x_T}_{\sim p_{T+1}^{\text{pre}}(\cdot)} \longrightarrow \underbrace{x_{T-1}}_{\sim p_T^{\text{pre}}(\cdot | x_T)} \longrightarrow \dots \longrightarrow \underbrace{x_2}_{\sim p_3^{\text{pre}}(\cdot | x_3)} \longrightarrow \underbrace{x_1}_{\sim p_2^{\text{pre}}(\cdot | x_2)} \longrightarrow \underbrace{x_0}_{\sim p_1^{\text{pre}}(\cdot | x_1)}$$

■ PRETRAINED SAMPLE DISTRIBUTION

$$p^{\text{pre}}(x_0) = \int_{x_{1:T}} p^{\text{pre}}(x_{0:T}) d_{x_{1:T}} = \int_{x_{1:T}} \prod_{t=1}^T p_t^{\text{pre}}(x_{t-1} | x_t) p_{T+1}^{\text{pre}}(x_T) d_{x_{1:T}}$$

Fine-tuning model

Fine-tuning diffusion model: $p_t(x_{t-1} | x_t)$ for $t = T, \dots, 2, 1$

Trajectory distribution: $p(x_{0:T}) = \prod_{t=T}^1 p_t(x_{t-1} | x_t) p_{T+1}(x_T)$

Target distribution: $p(x_0) = \int_{x_{1:T}} p(x_{0:T}) d_{x_{1:T}}$

Reward model: $r(x_0)$ for a sample x_0

■ REWARD MAXIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)} [r(x_0)]$$

■ PRETRAINED MODEL RETAINING

$$\underset{p}{\text{minimize}} \quad D_{\text{KL}}(p(\cdot) \parallel p^{\text{pre}}(\cdot))$$

Fine-tuning w/ point-wise KL-regularization

Trajectory distribution: $p(x_{0:T}) = \prod_{t=T}^1 p_t(x_{t-1} | x_t) p_{T+1}(x_T)$

Target distribution: $p(x_0) = \int_{x_{1:T}} p(x_{0:T}) d_{x_{1:T}}$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)} [r(x_0)] - \beta D_{\text{KL}}(p(\cdot) \| p^{\text{pre}}(\cdot))$$

$\beta > 0$ – tradeoff parameter

★ optimal model p^*

$$p^*(x_0) \propto p^{\text{pre}}(x_0) e^{r(x_0)/\beta}$$

Intractable path marginalization

Relaxation of point-wise KL divergence

■ UPPER BOUND OF POINT-WISE KL DIVERGENCE

$$\begin{aligned} D_{\text{KL}}(p(\cdot) \parallel p^{\text{pre}}(\cdot)) &= \mathbb{E}_{x_0 \sim p(\cdot)} \left[\log \frac{p(x_0)}{p^{\text{pre}}(x_0)} \right] \\ &\leq \mathbb{E}_{x_{0:T} \sim p(\cdot)} \left[\log \frac{p(x_{0:T})}{p^{\text{pre}}(x_{0:T})} \right] \end{aligned}$$

$$D_{\text{KL}}(P_X \parallel Q_X) \leq D_{\text{KL}}(P_{X,Y} \parallel Q_{X,Y}) - \text{monotonicity}$$

Tractable path expectation

Path-wise KL divergence

■ PATH DECOMPOSITION

$$\begin{aligned} & \mathbb{E}_{x_{0:T} \sim p(\cdot)} \left[\log \frac{p(x_{0:T})}{p^{\text{pre}}(x_{0:T})} \right] \\ &= \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t), t=1, \dots, T, T+1} \left[\log \frac{p(x_{0:T})}{p^{\text{pre}}(x_{0:T})} \right] \\ &= \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t), t=1, \dots, T, T+1} \left[\log \prod_{t=T+1}^1 \frac{p_t(x_{t-1} | x_t)}{p_t^{\text{pre}}(x_{t-1} | x_t)} \right] \\ &= \sum_{t=T+1}^1 \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t)} \left[\log \frac{p_t(x_{t-1} | x_t)}{p_t^{\text{pre}}(x_{t-1} | x_t)} \right] \\ &= \mathbb{E}_{x_{1:T} \sim p(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t) \parallel p_t^{\text{pre}}(\cdot | x_t)) \right] \end{aligned}$$

$p_{T+1}^{\text{pre}}(\cdot | x_{T+1}) = p_{T+1}^{\text{pre}}(\cdot | x_{T+1})$ – fixed Gaussian

Fine-tuning w/ path-wise KL-divergence

Trajectory distribution: $p(x_{0:T}) = \prod_{t=1}^T p_t(x_{t-1} | x_t) p_{T+1}(x_T)$

Target distribution: $p(x_0) = \int_{x_{1:T}} p(x_{0:T}) d_{x_{1:T}}$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta \mathbb{E}_{x_{0:T} \sim p(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right]$$

$\beta > 0$ – tradeoff parameter

Finite-horizon policy optimization

MDP model

■ MARKOV CHAIN

$$\underbrace{x_T}_{\sim p_{T+1}(\cdot)} \longrightarrow \underbrace{x_{T-1}}_{\sim p_T(\cdot | x_T)} \longrightarrow \dots \longrightarrow \underbrace{x_2}_{\sim p_3(\cdot | x_3)} \longrightarrow \underbrace{x_1}_{\sim p_2(\cdot | x_2)} \longrightarrow \underbrace{x_0}_{\sim p_1(\cdot | x_1)}$$

- ★ **continuous** state and action spaces $S = A = X \subset \mathbb{R}^d$
- ★ initial state distribution $s_0 = x_T \sim p_{T+1}(\cdot)$, $s_t = x_{T-t}$ for $t = 1, 2, \dots, T - 1$
- ★ policy at time t is $a_t \sim \pi_t(\cdot | s_t) = p_{T-t}(\cdot | x_{T-t})$
- ★ **deterministic** transition to next state $s_{t+1} = a_t$
- ★ zero reward $r(s_t, a_t) = 0$ for $t = 0, 1, \dots, T - 1$

Computation of an optimal policy via **dynamic programming**

Optimal policy

■ FINE-TUNING W/ PATH-WISE KL-REGULARIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta \mathbb{E}_{x_{0:T} \sim p(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t)) \right]$$

$\beta > 0$ – tradeoff parameter

★ optimal policy $p^*(x_{0:T}) = \prod_{t=T}^1 p_t^*(x_{t-1} | x_t) p_{T+1}(x_T)$

$$p_t^*(x_{t-1} | x_t) = \underset{p_t}{\text{argmax}} \quad \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t), x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) - \beta \sum_{k=t-1}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) \right] - \beta D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t))$$

■ OPTIMAL VALUE FUNCTION

$$v_{t-1}(x_{t-1}) = \mathbb{E}_{x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) - \beta \sum_{k=t-1}^1 D_{\text{KL}}(p_k^*(\cdot | x_k) \| p_k^{\text{pre}}(\cdot | x_k)) \right]$$
$$v_0(x_0) = r(x_0)$$

■ STAGE-WISE POLICY OPTIMIZATION

$$p_t^*(x_{t-1} | x_t) = \underset{p_t}{\operatorname{argmax}} \mathbb{E}_{x_{t-1} \sim p_t(\cdot | x_t)} [v_{t-1}(x_{t-1})] - \beta D_{\text{KL}}(p_t(\cdot | x_t) \| p_t^{\text{pre}}(\cdot | x_t))$$

★ optimal policy at time t is $p_t^*(x_{t-1} | x_t)$

$$p_t^*(\cdot | x_t) \propto p_t^{\text{pre}}(\cdot | x_t) e^{v_{t-1}(\cdot)/\beta}$$

Fine-tuning in parameter space

Trajectory distribution: $p_{\theta}(x_{0:T}) = \prod_{t=1}^T p_t(x_{t-1} | x_t; \theta) p_{T+1}(x_T)$ w/ parameter θ

Reward model: $r(x_0)$ for a sample x_0

■ PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p_{\theta}(\cdot)}[r(x_0)] - \beta \mathbb{E}_{x_{0:T} \sim p_{\theta}(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t; \theta) \| p_t^{\text{pre}}(\cdot | x_t; \theta^{\text{pre}})) \right]$$

$\beta > 0$ – tradeoff parameter

Policy gradient: REINFORCE

■ POLICY GRADIENT

$$\begin{aligned} & \nabla_{\theta} \mathbb{E}_{x_0 \sim p_{\theta}(\cdot)} [r(x_0)] - \beta \nabla_{\theta} \mathbb{E}_{x_{0:T} \sim p_{\theta}(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t(\cdot | x_t; \theta) \| p_t^{\text{pre}}(\cdot | x_t; \theta^{\text{pre}})) \right] \\ &= \nabla_{\theta} \mathbb{E}_{x_T \sim p_{T+1}(\cdot), x_{T-1} \sim p_T(\cdot | x_T; \theta), \dots, x_0 \sim p_1(\cdot | x_1; \theta)} [r(x_0)] \\ & \quad - \beta \nabla_{\theta} \mathbb{E}_{x_T \sim p_{T+1}(\cdot), x_{T-1} \sim p_T(\cdot | x_T; \theta), \dots, x_0 \sim p_1(\cdot | x_1; \theta)} \left[\sum_{t=T}^1 \frac{\|\mu_{\theta}(x_t, t) - \mu_{\theta^{\text{pre}}}(x_t, t)\|^2}{2\sigma^2(t)} \right] \\ &= \mathbb{E}_{x_T \sim p_{T+1}(\cdot), x_{T-1} \sim p_T(\cdot | x_T; \theta), \dots, x_0 \sim p_1(\cdot | x_1; \theta)} \left[\sum_{t=T}^1 \tilde{r}(x_{0:T}) \nabla_{\theta} \log p_t(x_{t-1} | x_t; \theta) \right] \\ & \quad - \beta \mathbb{E}_{x_T \sim p_{T+1}(\cdot), x_{T-1} \sim p_T(\cdot | x_T; \theta), \dots, x_0 \sim p_1(\cdot | x_1; \theta)} \left[\sum_{t=T}^1 \frac{\nabla_{\theta} \|\mu_{\theta}(x_t, t) - \mu_{\theta^{\text{pre}}}(x_t, t)\|^2}{2\sigma^2(t)} \right] \end{aligned}$$

Easy to get an unbiased estimate

Summary

■ RECAP

■ ALIGNMENT OF DM

Downstream requirements

■ REWARD-BASED ALIGNMENT

Point-wise KL divergence, path-wise KL divergence, stage-wise policy optimization, policy gradient